

VisionSort: Automated Multi-Category Waste Classification Pipeline

PROJECT REPORT

Submitted By

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Project Title

VisionSort: Automated Multi-Category Waste Classification Pipeline

Technologies Used

- Python
- TensorFlow / Keras
- FastAPI
- MongoDB
- HTML
- CSS
- JavaScript
- Chart.js

PROJECT FOLDER STRUCTURE

```
VisionSort/  
|— backend/  
|— frontend/  
|— dataset/  
|— model/  
|— screenshots/  
|— report/  
|— requirements.txt  
|— README.md
```

ABSTRACT

Waste segregation is one of the most important processes in modern recycling industries. However, most recycling centers still depend heavily on manual sorting, resulting in contamination, slower processing, higher operational costs, and reduced recycling efficiency. Human workers often struggle to accurately distinguish between plastics, metals, cardboard, paper, glass, and non-recyclable waste during continuous sorting operations.

VisionSort is an Artificial Intelligence-powered waste classification system developed to automate this process using Computer Vision and Deep Learning. The system employs a custom Convolutional Neural Network (CNN) trained on a labeled waste image dataset to classify six categories of waste materials. The trained model is deployed through a FastAPI backend, enabling real-time image classification through REST APIs.

Prediction history, confidence scores, and waste classification statistics are stored in MongoDB, while an industrial-style dashboard built using HTML, CSS, JavaScript, and Chart.js visualizes live classification results, confidence levels, analytics, and system information.

The developed solution demonstrates how AI can significantly improve recycling efficiency while reducing contamination and human effort. The architecture is modular and can be extended to conveyor belt systems, IoT-enabled recycling plants, and smart city waste management solutions.

1. INTRODUCTION

Rapid urbanization and industrial growth have dramatically increased the amount of municipal solid waste generated worldwide. Proper segregation of recyclable materials is essential for effective recycling and environmental sustainability. However, traditional waste sorting is still largely dependent on manual labor.

Manual segregation introduces several limitations including inconsistent classification, slower processing speed, worker fatigue, and increased contamination of recyclable materials. These challenges reduce the efficiency of recycling plants and increase operational costs.

Artificial Intelligence, particularly Deep Learning, provides an opportunity to automate waste segregation through image classification techniques. By training neural networks on labeled waste images, computers can accurately identify different material categories without human intervention.

VisionSort was developed as a complete AI-powered waste classification platform combining Deep Learning, Backend APIs, Database Management, and Interactive Web Technologies. Unlike standalone machine learning models, VisionSort integrates all software engineering components required for deployment into a single industrial-style application.

The project demonstrates the practical application of Artificial Intelligence in smart waste management while following modern software architecture principles.

2. PROBLEM STATEMENT

Material recycling centers suffer from massive contamination because manual sorting lines fail to accurately separate plastics, metals, cardboard, paper, glass, and other waste materials. Human operators experience fatigue, reduced consistency, and slower sorting speeds during continuous waste processing.

As contamination increases, recyclable materials lose value and significant quantities of reusable waste end up in landfills. Existing manual systems therefore increase operational costs while reducing environmental sustainability.

The objective of VisionSort is to automate waste classification using Artificial Intelligence by accurately identifying multiple waste categories from images. The system provides confidence scores, disposal recommendations, prediction analytics, and centralized monitoring through an industrial dashboard.

3. OBJECTIVES

The major objectives of VisionSort include:

- Develop an AI-based waste classification system using Deep Learning.
 - Train a custom CNN capable of identifying multiple waste categories.
 - Build a FastAPI backend for real-time image classification.
 - Store prediction history and analytics using MongoDB.
 - Design an industrial dashboard displaying live predictions and confidence scores.
 - Reduce manual waste segregation errors.
 - Improve recycling efficiency through automated material identification.
 - Demonstrate practical deployment of Artificial Intelligence in sustainable waste management.
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4. EXISTING SYSTEM

Current recycling centers primarily rely on manual sorting processes. Workers visually inspect waste items and separate them into different recycling bins. Although this approach is simple, it introduces several limitations.

Limitations of Existing System

- High dependency on manual labor
- Slow processing speed
- Human classification errors
- Worker fatigue
- High contamination rates
- Increased operational costs
- Low scalability
- Poor analytics and monitoring

Manual sorting systems also lack real-time analytics and centralized monitoring capabilities, making it difficult for recycling plants to optimize their operations.

5. PROPOSED SYSTEM

VisionSort proposes an AI-powered automated waste classification pipeline capable of recognizing multiple waste categories through image analysis.

The proposed system consists of four major components:

AI Layer

A custom Convolutional Neural Network (CNN) trained using TensorFlow/Keras performs image classification across six waste categories.

Backend Layer

FastAPI provides REST APIs for image upload, model inference, prediction generation, and communication between frontend and AI model.

Database Layer

MongoDB stores prediction history, confidence values, timestamps, and analytics for future visualization.

Frontend Layer

A responsive HTML/CSS/JavaScript dashboard displays uploaded images, prediction results, confidence levels, analytics charts, and system information.

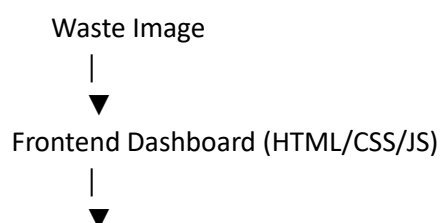
Advantages of Proposed System

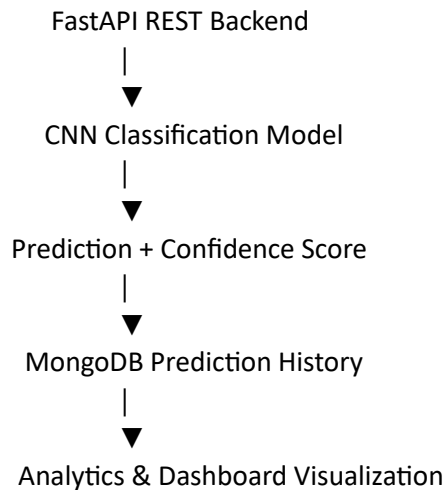
- Automated waste classification
- Reduced contamination
- Fast prediction
- Confidence score generation
- Industrial dashboard
- Prediction history
- Analytics visualization
- Easy scalability
- Cloud deployment ready
- Extensible for real-time video processing

6. ENGINEERING ARCHITECTURE

VisionSort follows a modular architecture that integrates Artificial Intelligence, Backend APIs, Database Management, and a Responsive User Interface into a unified waste classification system.

The architecture consists of four major layers:





Each layer performs a specific responsibility while maintaining loose coupling with the remaining components, making the system scalable and maintainable.

7. AI LAYER

The Artificial Intelligence layer forms the core of VisionSort. It is responsible for automatically identifying waste categories from uploaded images.

Unlike traditional rule-based systems, VisionSort utilizes a **Custom Convolutional Neural Network (CNN)** developed using TensorFlow/Keras. The CNN learns visual patterns from thousands of labeled waste images and predicts the category of newly uploaded images.

The AI pipeline consists of:

Image Acquisition

The user uploads an image through the dashboard.

Image Preprocessing

The uploaded image undergoes preprocessing before being passed to the CNN model.

Processing steps include:

- Image resizing to **224 × 224 pixels**
- RGB conversion
- Pixel normalization
- Tensor conversion

These preprocessing steps ensure consistency between training and inference.

CNN Model

The custom CNN contains multiple convolutional layers, activation functions, pooling layers, dense layers, and dropout regularization.

The model extracts:

- Texture features
- Edge features
- Shape information
- Material patterns

Finally, a Softmax classifier predicts the probability of each waste category.

Supported Waste Categories

The model classifies images into six categories:

- Cardboard
 - Glass
 - Metal
 - Paper
 - Plastic
 - Trash
-

Prediction Pipeline

Uploaded Image
|
Resize
|
Normalize
|
CNN Feature Extraction
|
Softmax Classification
|
Confidence Score
|
Final Prediction

Confidence Score

The CNN produces probability values for each class.

The class with the highest probability is selected as the predicted category.

Example:

Plastic : 90.81%

Glass : 4.52%

Paper : 2.40%

Metal : 1.20%

Trash : 0.70%

Cardboard : 0.37%

Highest probability = **Plastic**

8. BACKEND & DATABASE

VisionSort uses **FastAPI** as the backend framework because of its high performance and lightweight architecture.

The backend is responsible for:

- Receiving uploaded images
- Preprocessing images
- Running CNN inference
- Returning prediction results
- Calculating confidence scores
- Generating disposal recommendations
- Managing prediction history
- Providing analytics endpoints

Backend Workflow

Image Upload



FastAPI Endpoint



Image Processing



CNN Prediction



Generate Response



Store in MongoDB



Return JSON Response

FastAPI APIs

The backend exposes multiple REST endpoints.

GET /

Checks whether the API server is running.

Returns:

- Server status
 - Model status
 - MongoDB connection status
 - Supported classes
-

POST /predict

Accepts an uploaded waste image.

Performs:

- Image preprocessing
- CNN inference
- Confidence calculation
- Risk assessment
- Disposal recommendation

Returns:

Predicted Class

Confidence

Risk Level

Inference Time

Timestamp

Disposal Recommendation

GET /history

Retrieves previously stored predictions from MongoDB.

Displays:

- Prediction history
 - Timestamp
 - Confidence
 - Waste category
-

GET /metrics

Provides analytics including:

- Total predictions
 - Waste distribution
 - Prediction statistics
-

MongoDB Integration

MongoDB stores prediction records for future analytics.

Each document contains:

Filename

Predicted Class

Confidence

Risk Level

Inference Time

Timestamp

Disposal Recommendation

MongoDB enables:

- Prediction history
 - Dashboard analytics
 - Future reporting
 - Performance monitoring
-

9. FRONTEND DASHBOARD

The frontend dashboard provides an industrial-style interface that allows users to interact with the AI model.

The dashboard is developed using:

- HTML5
 - CSS3
 - JavaScript
 - Chart.js
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Dashboard Modules

Live Waste Upload

Allows users to upload waste images.

Prediction Panel

Displays:

- Predicted category
 - Confidence score
 - Risk level
 - Disposal recommendation
 - Inference time
-

Analytics Dashboard

Displays:

- Total predictions
- Average confidence

- Waste category distribution
 - Doughnut chart
-

Prediction History

Maintains previously classified waste images.

System Information

Displays:

- AI Model
 - Backend
 - Database
 - Dataset
 - Number of Classes
 - System Status
-

AI Model Performance

Displays:

- Training Accuracy
 - Validation Accuracy
 - Number of Classes
 - Input Resolution
-

10. SYSTEM WORKFLOW

The operational workflow of VisionSort is illustrated below.

User Uploads Waste Image

↓

Frontend Dashboard

↓

FastAPI Backend

↓

Image Preprocessing



CNN Classification



Confidence Score Calculation



Prediction Generated



MongoDB Storage



Analytics Dashboard Updated



Prediction Displayed to User

11. DATASET

VisionSort uses the **TrashNet dataset**, a publicly available benchmark dataset for waste image classification.

Dataset Statistics

Parameter	Value
Total Images	~2500+
Categories	6
Image Format	JPG
Input Size	224 × 224
Model Type	CNN

Dataset Classes

- Cardboard

- Glass
 - Metal
 - Paper
 - Plastic
 - Trash
-

Data Preprocessing

Before training, all images undergo preprocessing:

- Image resizing
- Pixel normalization
- RGB conversion
- Dataset splitting into Training and Validation sets

These preprocessing steps improve training consistency and model convergence.

12. MODEL TRAINING

The Artificial Intelligence model used in VisionSort is a Custom Convolutional Neural Network (CNN) developed using TensorFlow/Keras. The objective of the model is to classify waste images into six predefined categories.

Training Process

The model was trained using the following steps:

Step 1 – Dataset Preparation

The TrashNet dataset was organized into six folders representing different waste categories.

Step 2 – Image Preprocessing

Before training, all images were preprocessed using:

- Image resizing to **224 × 224 pixels**
- RGB color conversion
- Pixel normalization (0–1 range)
- Dataset split into training and validation sets

Step 3 – CNN Architecture

The implemented CNN consists of:

- Convolution Layers
- ReLU Activation
- Max Pooling Layers

- Flatten Layer
- Fully Connected Dense Layers
- Dropout Layer
- Softmax Output Layer

The Softmax layer generates probability scores for all six waste categories.

Step 4 – Model Training

The model was trained using TensorFlow/Keras with multiple epochs until satisfactory classification performance was achieved.

The trained model was exported as:

waste_classifier.keras

and loaded by the FastAPI backend during inference.

13. EXPERIMENTAL RESULTS

The developed VisionSort system successfully classifies uploaded waste images using the trained CNN model.

Sample Prediction Output

Parameter	Output
Predicted Class	Plastic
Confidence	90.81%
Risk Level	LOW
Inference Time	~900 ms
Recommendation	Plastic Recycling

Model Performance

Metric	Value
Training Accuracy	~84%
Validation Accuracy	~64%
Number of Classes	6
Input Resolution	224 × 224

The dashboard also provides:

- Prediction history
 - Confidence visualization
 - Doughnut chart analytics
 - Average confidence
 - System information
 - Model performance summary
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14. DASHBOARD IMPLEMENTATION

A responsive industrial-style dashboard was developed using HTML, CSS, and JavaScript.

The dashboard provides an intuitive interface for users to upload waste images and visualize AI predictions.

Major Dashboard Components

1. Image Upload Panel

Allows users to upload waste images for AI analysis.

2. Prediction Panel

Displays:

- Predicted class
- Confidence score
- Risk level
- Disposal recommendation
- Inference time

3. Analytics Panel

Displays:

- Total predictions
- Average confidence
- Waste distribution
- Doughnut chart

4. Prediction History

Displays previously classified waste images stored in MongoDB.

5. System Information

Displays:

- AI Model
- Backend
- Database
- Dataset
- Number of Classes
- System Status

6. Model Performance

Displays:

- Training Accuracy
- Validation Accuracy
- Waste Classes
- Input Resolution

15. ADVANTAGES

VisionSort provides several advantages over traditional manual waste sorting systems.

- Automated waste classification
- Faster processing compared to manual sorting
- Reduced contamination in recycling
- Improved recycling efficiency
- Confidence score generation
- Real-time prediction
- Industrial dashboard for monitoring
- Prediction history storage
- Analytics visualization
- Scalable architecture
- Easy integration with cloud platforms
- Extensible for industrial automation

16. LIMITATIONS

Although VisionSort performs well for image-based waste classification, certain limitations remain.

- The model currently classifies uploaded images rather than continuous live video streams.

- Prediction accuracy depends on image quality and lighting conditions.
- Performance can be improved with larger and more diverse datasets.
- The current CNN architecture can be enhanced using transfer learning techniques such as MobileNetV2 or EfficientNet.

17. FUTURE SCOPE

VisionSort can be extended into a complete industrial waste management platform.

Future enhancements include:

- Real-time conveyor belt waste classification
- Live camera stream processing
- YOLOv8 object detection for multiple waste objects
- Transfer learning using MobileNetV2/EfficientNet
- IoT integration with smart recycling bins
- Cloud deployment using Docker and Kubernetes
- Mobile application for waste classification
- Admin dashboard with role-based authentication
- Automatic recycling statistics and reporting
- AI-powered waste volume forecasting

18. CONCLUSION

VisionSort demonstrates the practical application of Artificial Intelligence in modern waste management systems. By integrating Deep Learning, FastAPI, MongoDB, and an interactive web dashboard, the project provides an end-to-end solution for automated waste classification.

The system successfully identifies six categories of waste, generates confidence scores, stores prediction history, and visualizes analytics through a responsive industrial dashboard. The modular architecture allows the project to be extended for real-world industrial applications, including conveyor belt automation and smart recycling systems.

Overall, VisionSort reduces dependence on manual waste segregation, improves recycling efficiency, and contributes toward sustainable waste management practices.

19. REFERENCES

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